

Values-Led Problem Statement

Optimizing RAG Document Retrieval for Agronomic Advice

Team Name	RasDashen
Selected Project	Optimizing RAG Document Retrieval for Agronomic Advice
Challenge	Challenge 1 — Values-Led Problem Statement
Team Members	Olumide Adekunle, Misturah, Sabona, Ogwuche Moses

The Problem

Smallholder farmers across many African regions rely heavily on agricultural extension officers for timely, locally relevant advice on crop diseases, pests, soil management, and climate adaptation. Extension services are chronically understaffed relative to the number of farmers who need support, and the written extension material that does exist — government and NGO guidance, technical bulletins, and reference documents — is scattered across many sources. This makes it slow for even a knowledgeable extension officer to locate the single most relevant passage for a farmer's specific question, and effectively out of reach for a farmer trying to find an answer on their own.

Who This Affects

The people most affected are smallholder farmers, who bear the direct consequences (in yield and income) of delayed or unclear guidance, and the extension officers and organisations who are stretched thin trying to serve them. Both groups are underserved by the current state of scattered, hard-to-search reference material — not because the knowledge doesn't exist, but because finding the right piece of it quickly is difficult.

Our Values-Led Aim

Our team's aim is to reduce this bottleneck: build a document retrieval system that surfaces the most relevant agronomic extension material for a farmer's question, so that farmers and the extension workers who support them can get trustworthy, source-grounded answers faster. Three values guide this work:

Accuracy over fluency — a retrieval system that returns the wrong document can lead to real crop or livelihood loss, so precision matters more than eloquence.

Transparency — every answer is traceable to a real, human-authored source document, not a generated claim, so it can be independently verified by a farmer, officer, or reviewer.

Accessibility — the system must run on modest, low-cost compute (no GPU, CPU-only retrieval), so it can realistically be deployed in low-resource settings rather than requiring infrastructure most extension organisations don't have.

What Success Looks Like

A working retrieval system that, given a farmer's question in plain language, returns the top-5 most relevant documents from an agronomic knowledge base with measurably better relevance (by nDCG@5) than a simple keyword-matching baseline — while remaining transparent about where its knowledge base has gaps, so it is deployed and understood honestly rather than oversold.

Prepared for Challenge 1 (Values-Led Problem Statement) submission.